

Log set L_s with resource information

Prediction phase

Step 1: Train and select the best model LR

↓ LR

Step 2: Predict the distinct resource number n

Log L' lacks resource information

Extraction phase

Extract Domain Distributions:

- D1:** Handover count frequency per case
- D2:** Distinct resource count frequency per case
- D3:** Event trigger frequency per resource
- D4:** Case involvement frequency per resource

Train $LRC1$

Train $LRC2$

Inference phase

Step 1: handover detection (using **D1**, $LRC1$)

↓ L_h' with detected handovers

Step 2: resource labeling per case (using **D2**, $LRC2$)

↓ L_o' with case-level resource labels

Step 3: resource assignment for the log (using **D3**, **D4**, n)

Log L'' with inferred resource information